

lampe de Wut permet de voir plus facilement les Δ de pigmentation
(à la naissance : Δ pigmentation peu visible car pas d'exposition UV in-utéro)

!! différencier :

- hypopigmentation = spectre
- dépigmentation = macule blanche

White spots on the skin

-> macule blanche

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Objectives

- **Develop skills in observing and describing skin lesions**
- **Establish a differential diagnosis**
- **Recognize white spots that require no further evaluation**
- **Recognize white spots that may represent skin manifestations of a more complex problem**

Differential diagnosis

Congenital

- **Nevus depigmentosus or achromicus**
- **Nevus anemicus**
- **Pigmentary mosaicism**
- **Tuberous sclerosis** -> CAVE : épilepsie
- **Piebaldism**

Differential diagnosis

Acquired

- **Post-inflammatory hypopigmentation**
- **Vitiligo**
- **Pityriasis alba** -> hypopigmentation, signe discret de dermatite
- **Pityriasis versicolor**
- **Cutaneous T-cell lymphoma** rare
- **Leprosy** rare

Diagnosis ?



nouvelles lésions continues à apparaitre
Dx posé par les macules dépigmentée (si ce n'est pas l'épilepsie)

Diagnostic evaluation of Tuberous sclerosis

- **CT scan/MRI of brain, electroencephalogram**
- **Echocardiography**
- **Renal ultrasound**
- **Fundoscopy**
- **Bone survey**
- **Chest radiography**
- **Screen first-degree relatives**

Tuberous sclerosis

angiofibrome des paumettes, des replis
inguéales plus tardives

-> Sclérose tubéreuse de Bourdeville

**Development of hamartomas of the skin, brain, eye,
heart, kidneys, lungs, and bone**

- **Autosomal dominant**
- **1:10' 000; M=F; all races**
- **75% spontaneous mutations**
- **Variable expressivity**

Tuberous sclerosis

- **Hypopigmented macules***
and confetti lesions+
 - at birth or soon after (90%)
- **Fibrous forehead plaque***
 - at birth (25 %)
- **Facial Angiofibromas***
 - age 2 to 6 years (80%)
- **Shagreen patch***
 - later childhood (20%)
- **Periungual fibromas***
 - after puberty (50%)
- **Café-au-lait macule+**
- **Gingival fibromas+**

Major features:*, Minor features:+

angiofibrome

=> donner des crèmes solaires pour éviter la mutation acquise (UV) sur le deuxième allèle ce qui ferait croître les angiofibromes cutanés (1ère mutation est congénitale)

Tuberous sclerosis

- **CNS** épilepsies !!
 - **Seizures+** (85%), mental retardation+ (60%), tumors
 - **Cortical tubers*, subependymal nodules* or astrocytomas***
- **Cardiac rhabdomyomas* (80%)**
- **Renal angiomyolipomas* (15%) and cysts+**
- **Retinal hamartomas* (50%) and achromic patches+**
- **Pulmonary lymphangiomyomatosis* and cysts+**
- **Bone cysts+**
- **Enamel pits+**
- **Hamartomatous rectal polyps+**

Major features:*, Minor features:+

Diagnosis ?



Nevus depigmentosus

-> juste un défaut de dépigmentation localisé

- **May be present at birth or appear during early infancy and persist unchanged lifelong**
- **Mostly solitary lesions**, but multiple and segmental lesions have been described

Diagnosis ?



Nevus anemicus

- **Developmental anomaly of cutaneous blood vessels**
- **Characterized by a circumscribed round or oval patch of pale skin (vasoconstricted patch of skin)**

Diagnosis ?



Vitiligo

maladie de dépigmentation la plus fréquente

faciale, ano-génitale, mains, articulations genoux, coude

-> zone de frictions = déstabilisation des mélanocytes

-> suiv de leurs destructions

-> maladie auto-immune des mélanocytes

- **Prevalence**
 - 1% of the population
 - 7-12% among first-degree relatives
 - 23% among monozygotic twins
- **Mean age in pediatric patients is 6.2 years**
- **Rarely congenital**

Vitiligo

- **Immune mediated destruction of melanocytes**
- **Associated autoimmune disorders** (thyroidite d'Haschimoto)
 - increased frequency in family members
 - 1% of pediatric patients with vitiligo
- **Triggers**
 - **trauma** (Koebner phenomenon)
 - **physical and emotional stress**

Vitiligo

- **Depigmented macules with well defined, frequently hyperpigmented, convex borders**
- **Sites of predilection**
 - Body folds, body orifices, over bony prominences
- **12% show leukotrichia or poliosis**
- **Subtypes of Vitiligo**
 - generalized, focal, segmental, acrofacial, mucosal, and universal
- **Halo nevi are frequently seen**
- **Wood's light exam can be helpful**

1/3 -> résolution spontanée

1/3 -> stable

1/3 -> aggravation

Treatment of Vitiligo

ttt efficace : immunosuppression active

- **Potent topical steroids** twice-daily
- Calcineurin inhibitor twice daily
- **Topic or systemic psoralen and UVA** -> immunosuppression cutanée
- Narrowband ultraviolet B light
- Autologous grafting of non-lesional epidermis to affected areas
- **Cosmetic camouflage**, quick-tan preparations
- The course of Vitiligo is unpredictable
- Repigmentation appears as small spots, reflecting the migration of melanocytes from the hair follicle

Diagnosis ?



Piebaldism

-> défaut typiquement situé sur la partie frontale, ventral et qui ne touche pas la partie postérieure
=> problème de migration des mélanocytes depuis la crête neurale (origine vertébrale)
(mut. du réc. C-kyt)

- **Autosomal dominant disorder**
- **Mutation in the kit proto-oncogene**
 - encodes a cell surface receptor for the stem cell/mast cell growth factor
- **Clinical manifestations can be explained by**
 - defective migration of melanoblast from the neural crest to the ventral midline
 - defect in the differentiation of melanoblasts to melanocytes

Piebaldism

Characterized by

- **Congenital patterned areas of persistent depigmentation and hypopigmentation**
 - typically islands of normal and increased pigmentation within hypomelanotic areas, and some times hyperpigmented borders
 - chin, anterior neck, anterior portion of the trunk and abdomen, and extremities
- **White forelock in 80-90% of cases**
 - Can include the eyebrows and eyelashes and extend to the root of the nose

Piebaldism

Treatment

- **Cosmetic masking of areas of leucoderma**
- **Vigorous sun protection**
- **Reepithelialization by grafting with autologous cultured epidermis after laser removal of epidermis**

Diagnosis ?



Pigmentary Mosaicism

survient au niveau kératinocytaires
-> défaut de migrations !
déficiência de la part des kératinocytes à
synthétiser des facteurs de prolifération des
mélanocytes

- **Heterogenous group of disorders in which hypopigmentation or hyperpigmentation following the lines of Blaschko (lines of ectodermal embryologic development)**
- **30% have associated abnormalities of the bones, eyes, and/ or central nervous system (especially when extensive pigmentary changes are present)**
- **Mosaic conditions tend not to be hereditary**

Diagnosis ?



Post-inflammatory hypopigmentation

-> plus important chez les personnes ++ pigmentées

- May be associated with a wide **variety of inflammatory dermatoses or infections**
- Hypopigmentation is caused when keratinocytes injured by the inflammatory process are temporarily unable to accept melanosomes from the melanocyte dendrites
- Skin lesion becomes more obvious after UV exposure
- No therapy is effective, but tends to improve with time

Diagnosis ?



Pityriasis alba

- ~~Cause is unknown~~ -> souvent observé chez les enfants atopique
- **Nonspecific dermatitis with residual postinflammatory hypopigmentation**
- **On face, neck, upper trunk, and proximal extremities**
- **In children 3 to 16 years of age**
- **Often in individuals with darker skin types**
- **Often noted after sun exposure**

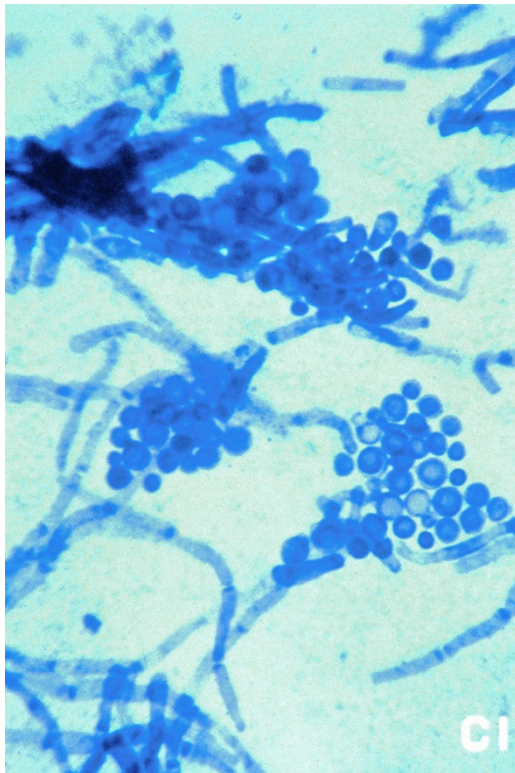
Treatment of Pityriasis alba

- **Mild topical corticosteroids or calcineurin inhibitors**
- **Sun protection of the site and surrounding area**
- **Emollients**
- **Repigmentation may take several months to years**

Diagnosis ?



Diagnosis ?



- Dx posé par la présence d'hydroxide de potassium

—

=> observation microscopique suffit à poser le Dx

- culture fongique n'est pas utile

Pityriasis versicolore

-> dire au patient que cela peut revenir

enfant pubertaire - adultes
lésions légers, squameuses, volontiers
hypopigmentées, se montrant dans les
gouttières antérieures et postérieures au niveau
du thorax

- Common superficial infection caused by **Malassezia furfur**

-> champignons

- Scaling, oval, hypopigmented or hyperpigmented macules
on upper part of the trunk, proximal arms, neck and face

- **Diagnosis is usually made clinically**

traitement préventif : se nettoyer
avant l'exposition au soleil
(shampooing au sélénium)

- **Lesions are more appeared after sun exposure**

-> quand il s'expose au soleil il
n'aura plus le champignon

- **Hypopigmentation results from the production of azelaic acid, which inhibits the dopa-tyrosinase reaction**

Treatment of Pityriasis versicolore

Topical-treatment

- **Selenium sulfide 2.5% shampoo**
 - Daily for 1-2 weeks then intermittently
- **Ketoconazole 2% shampoo**
 - Daily for 3 –5 days then intermittently
- **Terbinafine 1% spray 1-2x/ day for 1-2 weeks**
- **Topical antifungal creams**

Treatment of Pityriasis versicolore

If severe disease, recurrent disease or topical ttt failed

Oral-treatment

- **Ketoconazole single-dose 400mg followed by physical activity to promote secretion of the drug onto the skin via sweating, delay showering for 10-12 hours
Repeat ttt in a week**
- **Itraconazole single-dose 400mg or
7-day 200mg/day equally effective.**

Conclusions

Disorders of decreased pigmentation may be classified as

- 1. Genetic or developmentally controlled disorders, in which pigmentation tends to be abnormal from birth or early infancy**
- 2. Disorders associated with depigmentation or loss of previously existing melanin**